

JOHN LI

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EDUCATION

Carnegie Mellon University

Master's of Science in Mechanical Engineering Advanced Study

Concentration: Design and Manufacturing of Mechanical Systems

Pittsburgh, PA

Anticipated Graduation December 2027

Stony Brook University

Bachelor's Degree of Engineering in Mechanical Engineering

Honors: Dean's List (Top 20% of class) | University Scholar Program (Top 8% of the class) | GPA: 3.8

Stony Brook, NY

2022- 2026

ENGINEERING PROJECT & LEADERSHIP

SBU Student Launch Vehicle & Propulsion Subteam Lead | *Technical Leadership & Team Management* June 2025 - May 2026

- Managed full lifecycle development of two custom high-power competition airframes, engineering a Mars rover carrier and a precision altitude vehicle to achieve target apogees (1459 ft and 758 ft) within tight competition margins
- Conducted flight profile simulations using RockSim and MATLAB to optimize vehicle stability, verifying that aerodynamic center and center of gravity margins met structural constraints across the entire flight envelope
- Managed a \$1,400 team budget, prioritizing material procurement and fabrication costs to deliver two competition-ready airframes within constraints
- Served as Technical Lead to direct a 5-member design sub-team and coordinated full system integration across 15+ payload and recovery members, managing manufacturing schedules, design reviews, and safety compliance from PDR through launch day

PROFESSIONAL EXPERIENCES

Stantec

Commissioning Engineer Intern

New York, NY

Jun 2026 - Present

- Analyzing thermal/fluid system performance (chillers, AHUs, cooling towers, pumps) for a Manhattan facility retrofit, applying NYC TRM standards to evaluate equipment efficiency
- Quantifying energy savings and emissions (tCO₂e) using NYC TRM methodology to support data-driven retrofit specifications

Stony Brook University Materials and Mechanics Lab

NASA Space Grant Intern & Undergraduate Research Assistant

Stony Brook, NY

May 2025 - May 2026

- Conducted ASTM D6484 open-hole compression (OHC) testing on woven FRP laminates (hole sizes 3–6 mm) using a Shimadzu UTM and high-speed imaging to characterize load response and fracture process zone (FPZ) damage.
- Analyzed sub-scale failure modes across varying geometries, finding first-damage gross stress converging at 42–46 MPa — isolating a load-introduction-limited failure mode and informing future test protocol improvements.
- Continuing to investigate contributing variables: specimen length, grip clamping conditions, and hole machining quality

Brookhaven National Laboratory (BNL)

Mechanical Engineering Intern

Upton, NY

June 2024 - August 2024

- Utilized Creo and Ansys to model and structurally analyze new testing components for a facility supporting superconducting RF (SRF) cavity testing for the \$1.7-\$2.8 billion Electron-Ion Collider (EIC) accelerator program
- Performed onsite inspection and retrocommissioning assessment of weathered radiation-shielding components, previously stored outdoors for years, to evaluate structural viability for reuse in a new SRF cavity testing facility
- Prototyped a thermal test rig using additive manufacturing and embedded electronics to evaluate beryllium copper (BeCu) RF spring fingers, assessing whether the material could maintain sufficient contact force under cryogenic temperatures

TECHNICAL SKILLS

Softwares: Ansys, LabVIEW, MATLAB, SolidWorks, Creo, Inventor, Fusion 360, Arduino IDE

Technical: Project Management, Design & Prototyping, Manufacturing Processes (3D Printing & Manual & CNC Machining)
Finite Element Analysis (FEA), Circuit Building & Analysis, Microsoft Office, Google Suite